SIMATS ENGINEERING
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	**CSA1206**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL2-COMPARATIVE ANALYSIS**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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Department:	**CSE-AI**	**Date:**	**24-08-26**

ANNEXURE A
COMPARATIVE ANALYSIS

1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

*I **U.Prabhavathi/192472067** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:U.Prabhavathi

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Comparison of L1, L2 and L3 Cache

Cache memory is a small and high-speed memory located close to the processor. It stores frequently accessed instructions and data so that the CPU does not always need to access slower main memory.

L1 cache is the smallest and fastest cache. It has very low latency and very high bandwidth, so it can provide data to the CPU quickly. **L2 cache** is larger than L1 but slightly slower. It acts as a secondary cache when the required data is not found in L1. **L3 cache** is larger and is generally shared among multiple CPU cores. It has higher latency than L1 and L2 but helps reduce accesses to main memory.

Impact on performance: A higher cache hit rate reduces memory access time and improves processor throughput. Therefore, L1 has the greatest effect on immediate CPU performance, while L2 and L3 provide additional capacity and reduce costly DRAM accesses.

1. L1 vs L2 vs L3 Cache

Feature	L1 Cache	L2 Cache	L3 Cache
Location	Inside CPU core	Near/inside core	Shared among cores
Latency	Lowest	Low	Higher than L1/L2
Bandwidth	Highest	High	Lower
Size	Smallest	Medium	Largest
Throughput	Very high	High	Moderate
Purpose	Frequently used data/instructions	Backup for L1	Shared backup for L1/L2
Performance impact	Maximum	High	Moderate

Analysis: L1 provides the fastest access and highest bandwidth, so it has the greatest impact on CPU performance. L2 provides additional capacity with slightly higher latency. L3 is slower but reduces expensive main-memory accesses, improving overall processor throughput.

2. Comparison of SRAM and DRAM

SRAM and DRAM are two important types of semiconductor memory. SRAM (Static RAM) stores data using flip-flop circuits and does not require periodic refreshing. It provides very low latency and high bandwidth, making it suitable for CPU cache memory.

DRAM (Dynamic RAM) stores data using capacitors and requires periodic refreshing. It has higher latency than SRAM but provides greater storage density at a lower cost. Therefore, DRAM is commonly used as the computer's main memory.

Impact on performance: SRAM provides faster access and higher throughput, which is important for cache memory. DRAM provides large capacity at reasonable cost but is slower. The combination of SRAM cache and DRAM main memory creates an efficient memory hierarchy.

2. SRAM vs DRAM

Feature	SRAM	DRAM
Storage	Flip-flops	Capacitors
Latency	Very low	Higher
Bandwidth	High	Moderate to high
Throughput	Very high	High
Refresh	Not required	Required
Cost	Expensive	Cheaper
Density	Low	High
Typical use	CPU cache	Main memory

Analysis: SRAM is faster and provides higher throughput, making it suitable for L1, L2 and L3 **caches**. DRAM is slower but cheaper and denser, making it suitable for main memory**.**

3. Virtual Memory/Paging and Cache Memory

Virtual memory is a memory-management technique that allows a system to use secondary storage as an extension of physical RAM. Paging divides virtual memory and physical memory into fixed-size pages and frames. When a required page is not present in RAM, a page fault occurs and the system must retrieve it from storage.

Cache memory, on the other hand, is designed specifically to reduce the time required to access frequently used data. It is much faster than RAM and storage.

Impact on performance: Cache memory significantly reduces CPU access latency and increases system throughput. Virtual memory increases the amount of memory available to applications, but page faults can cause very high latency and temporarily reduce system performance.

3. Virtual Memory/Paging vs Cache Memory

Feature	Virtual Memory & Paging	Cache Memory
Purpose	Extends memory using storage	Speeds up CPU data access
Access latency	High	Very low
Storage	SSD/HDD + RAM	SRAM
Speed	Slow compared with cache	Very fast
Throughput	Lower	Very high
Replacement	Pages	Cache blocks
Main benefit	Provides larger address space	Reduces CPU memory-access time

Analysis: Cache memory improves processor throughput by keeping frequently used data close to the CPU. Virtual memory allows programs larger than physical RAM to execute, but page faults can introduce very high latency and reduce system throughput.

4. Comparison of NAND Flash and DRAM

NAND Flash is a non-volatile memory technology, meaning it retains data even when power is removed. It is widely used in SSDs, USB drives and memory cards. It provides high storage density and good sequential data-transfer performance, but its access and write latency are much higher than DRAM.

DRAM is volatile memory that loses its contents when power is removed. It provides much faster random access and higher throughput than NAND Flash. Because of its speed, DRAM is used as the main memory between CPU caches and secondary storage.

Suitability: DRAM is suitable for higher levels of the memory hierarchy where fast access is required. NAND Flash is more suitable for secondary storage where large capacity, low cost per bit and non-volatility are important.

4. NAND Flash vs DRAM

Feature	NAND Flash	DRAM
Latency	Higher	Lower
Bandwidth	Moderate to high	High
Throughput	Good for sequential operations	Very high for random access
Volatility	Non-volatile	Volatile
Write speed	Slower	Faster
Density	Very high	Lower
Typical use	SSD, storage	Main memory
Hierarchy level	Secondary storage	Main memory

Analysis: DRAM offers much lower latency and higher random-access performance, so it is placed closer to the processor. NAND Flash is slower but non-volatile, cheaper per bit, and highly dense, making it suitable for secondary storage.

5. Comparison of MRAM and RRAM

MRAM (Magnetoresistive RAM) stores information using magnetic states. It is non-volatile and provides fast access, high endurance and good energy efficiency. These properties make MRAM useful for embedded systems and applications requiring fast and persistent memory.

RRAM (Resistive RAM) stores information by changing the resistance of a material. It is also non-volatile and has the potential for high density and low power consumption. RRAM can provide fast access and good throughput, but practical performance can depend on endurance, device variation and manufacturing technology.

Performance: MRAM is attractive when fast access and high endurance are required, while RRAM is promising for high-density and energy-efficient memory systems. Both technologies are being investigated as alternatives or complements to conventional memory technologies.

5. MRAM vs RRAM

Feature	MRAM	RRAM
Technology	Magnetic memory	Resistive memory
Access time	Very fast	Very fast
Throughput	High	High
Energy efficiency	High	Very high potential
Non-volatile	Yes	Yes
Endurance	High	Moderate to high
Density	High	Potentially very high
Main advantage	Fast + high endurance	High density + low power

Analysis: MRAM provides fast access, high endurance, and good energy efficiency, making it suitable for cache-like and embedded applications. RRAM has strong potential for high-density, low-power memory, although factors such as endurance, variability, and manufacturing maturity can affect practical performance.